LEARNER GUIDE

Rise, run and two points

Straight Lines and Connected Representations. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

So far you have read the gradient straight from an equation like $y = 2x + 1$. This lesson does the opposite: you find the gradient from just two points on a line. You build a right-angled 'rise and run' triangle between the two points — the run is how far you go across (the change in x) and the rise is how far you go up or down (the change in y). The gradient is then rise divided by run. You will drag two points to build your own triangle, see that any two points on the same line give the same gradient, spot the two classic slips, and find real rates from two readings. The question you are exploring is: how can two points reveal the gradient without showing the equation?

What you are learning to notice

- That the run is the change in x (how far across) and the rise is the change in y (how far up or down).
- That the gradient is rise ÷ run — always in that order.
- That any two points on the same line give the same gradient — a wider triangle has a bigger rise AND a bigger run.
- That when the line falls, the rise is negative, so the gradient is negative.

Moving through it, one step at a time

In Same line, different triangle, look at two triangles on one line and check that rise ÷ run comes out the same for both. In the Gradient Triangle Builder, drag points A and B and read the rise, the run and the gradient — try to make a gradient of exactly 1, then something steeper. In Right or slip?, decide whether the working is correct or a slip (watch for run ÷ rise by mistake, or a lost minus sign). Then find a real rate from two readings, including one that goes down. There is no rush; use the coordinate grid on the Scratchpad to plot two points and draw the triangle yourself.

A gentle reminder
You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (reading coordinates, subtracting to find a difference (including negatives), and treating a fraction as a division), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.

- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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